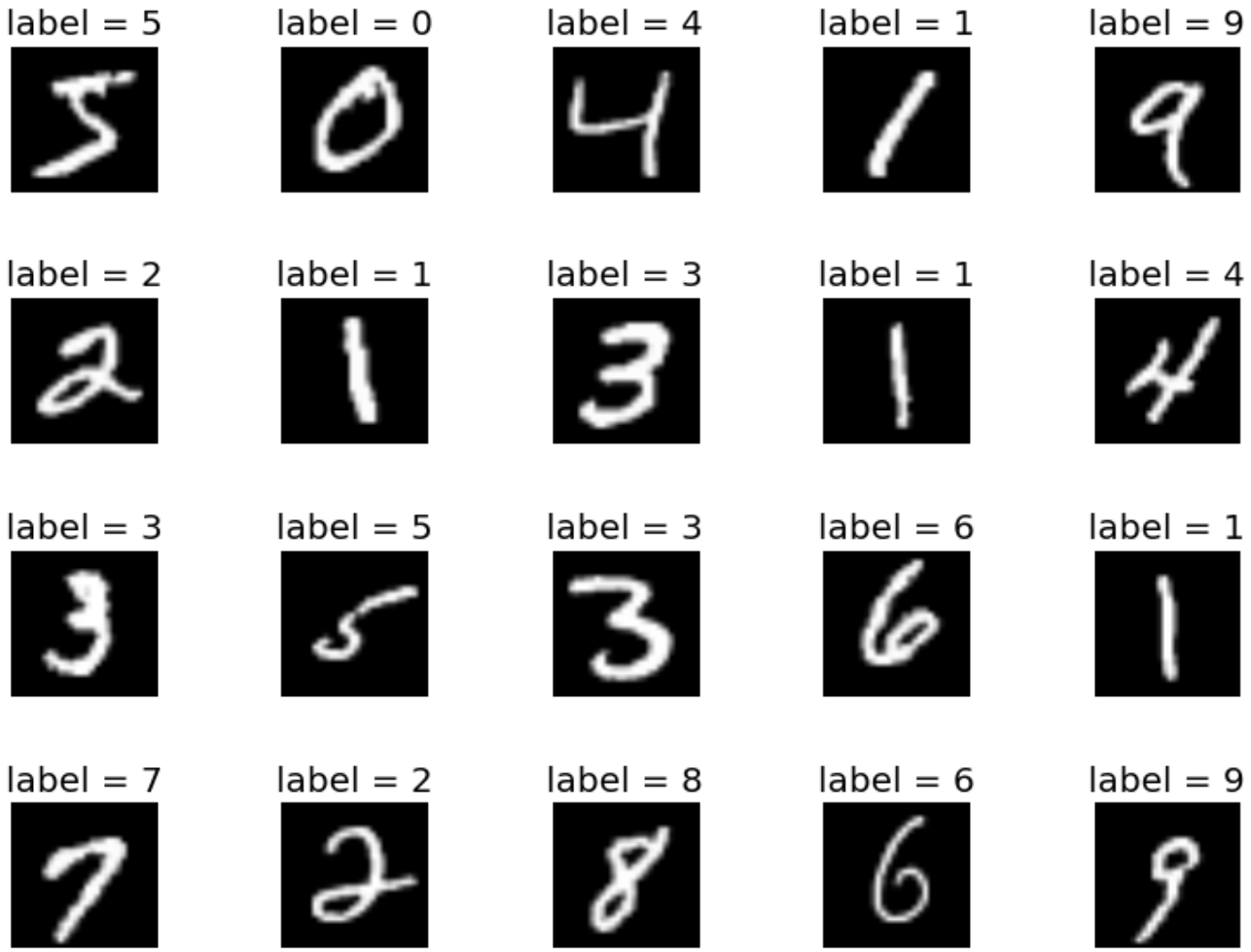
Formation 9 :

Le Deep Learning, Comment ca marche ?

KRYPTOSPHERE®

Quelques Rappels

Dataset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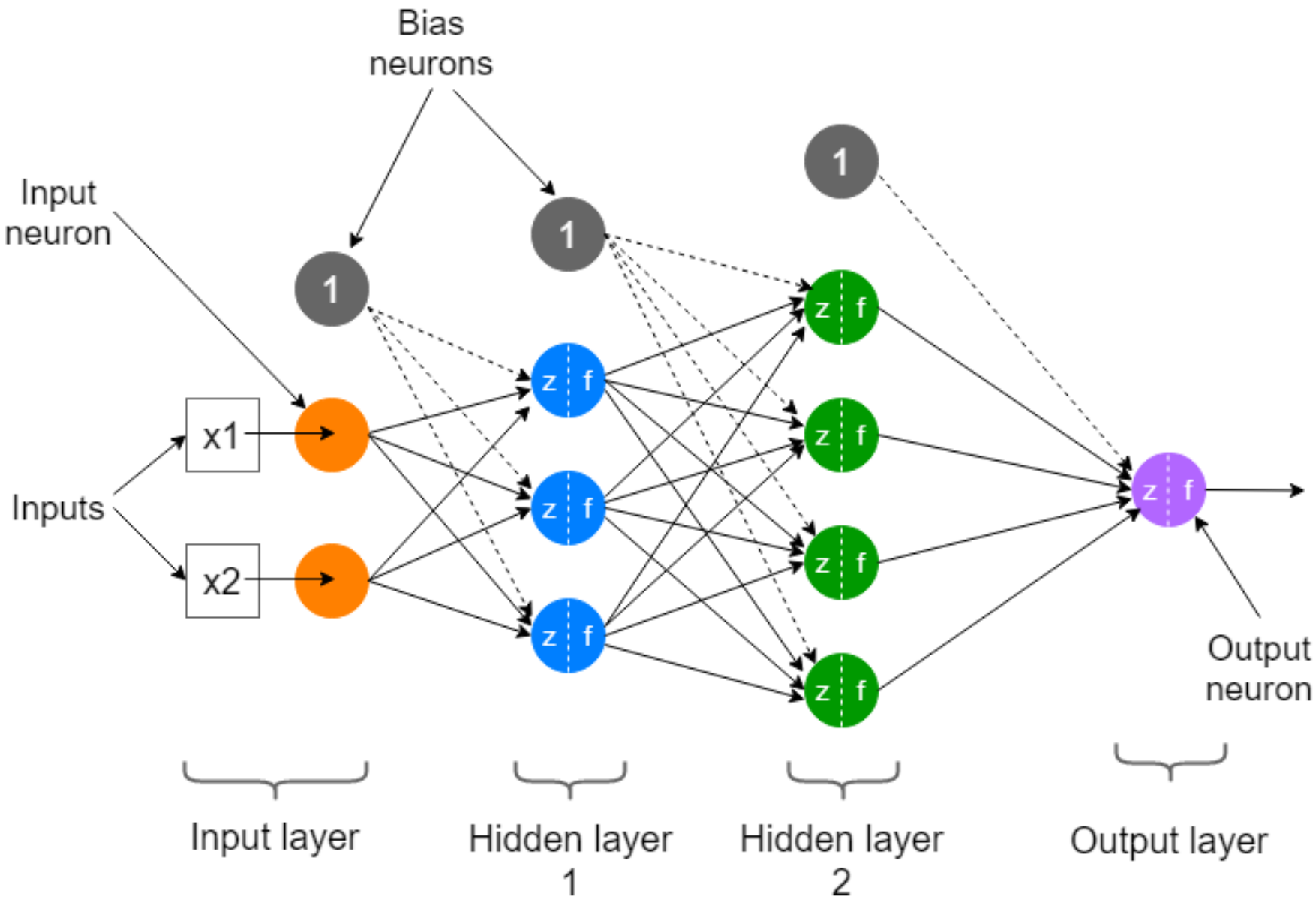
- Réseau de Neurone

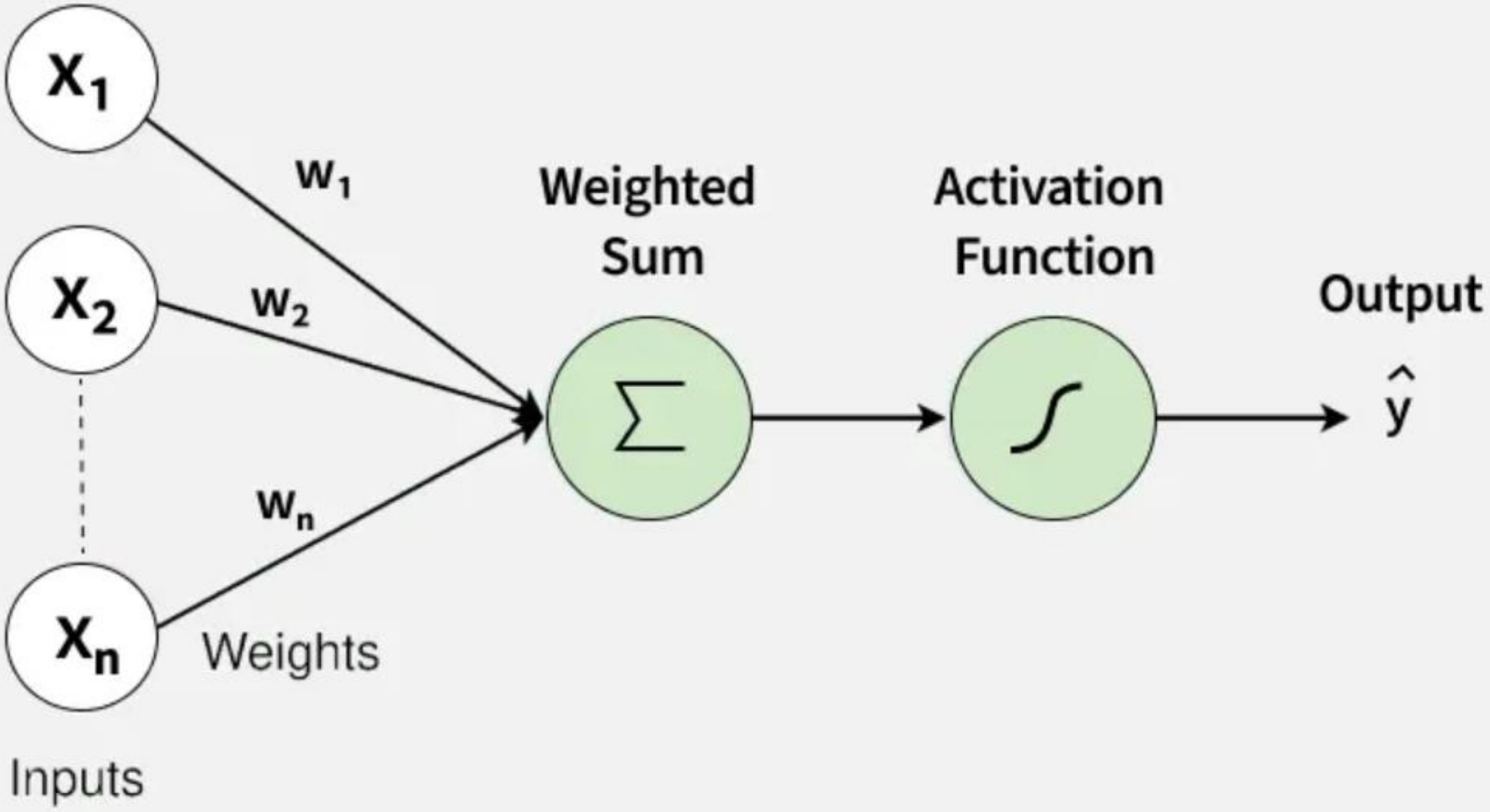


Comment on entraine un Réseau de Neurones ?

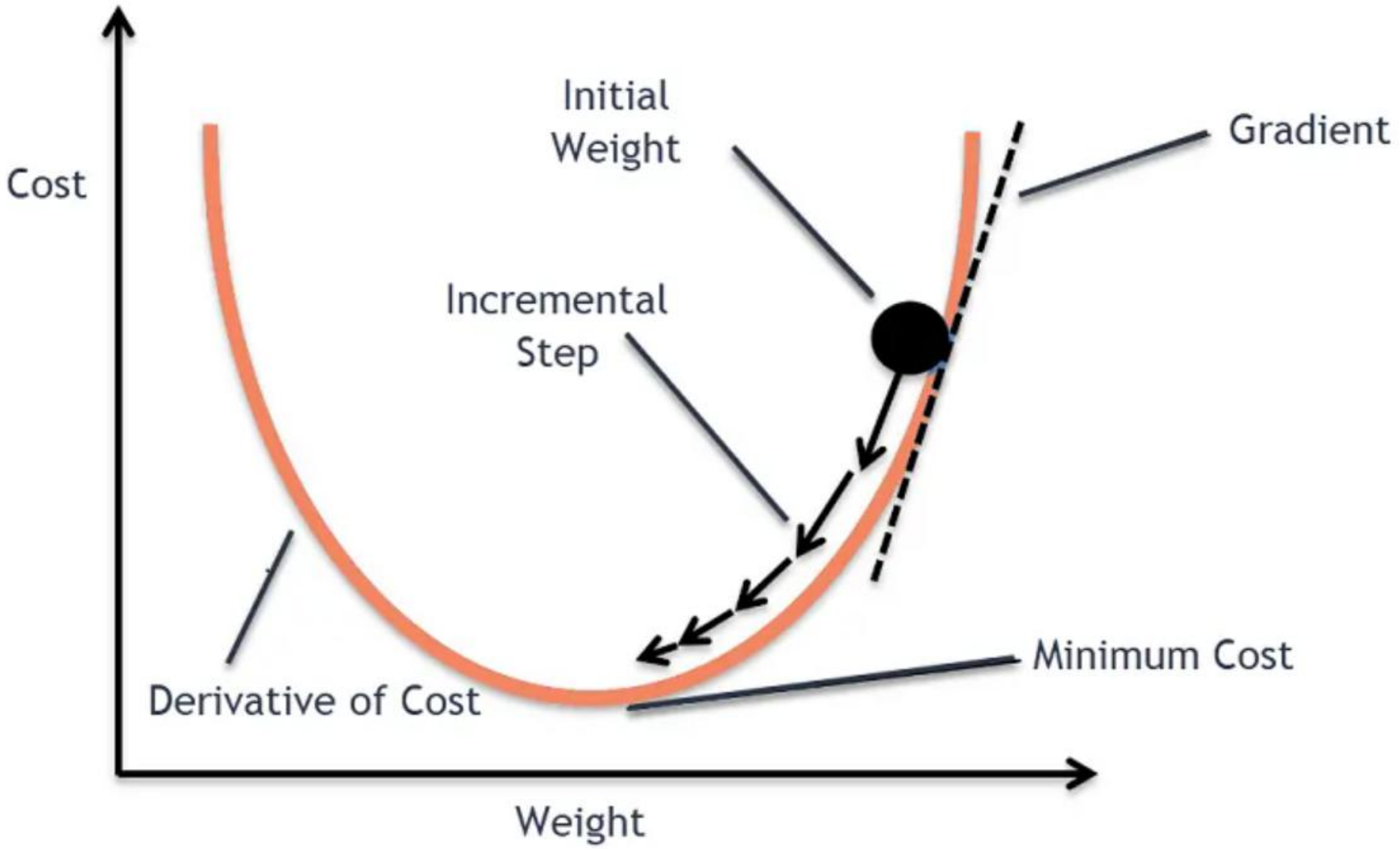
Choisir une fonction de perte : évaluer une prédiction

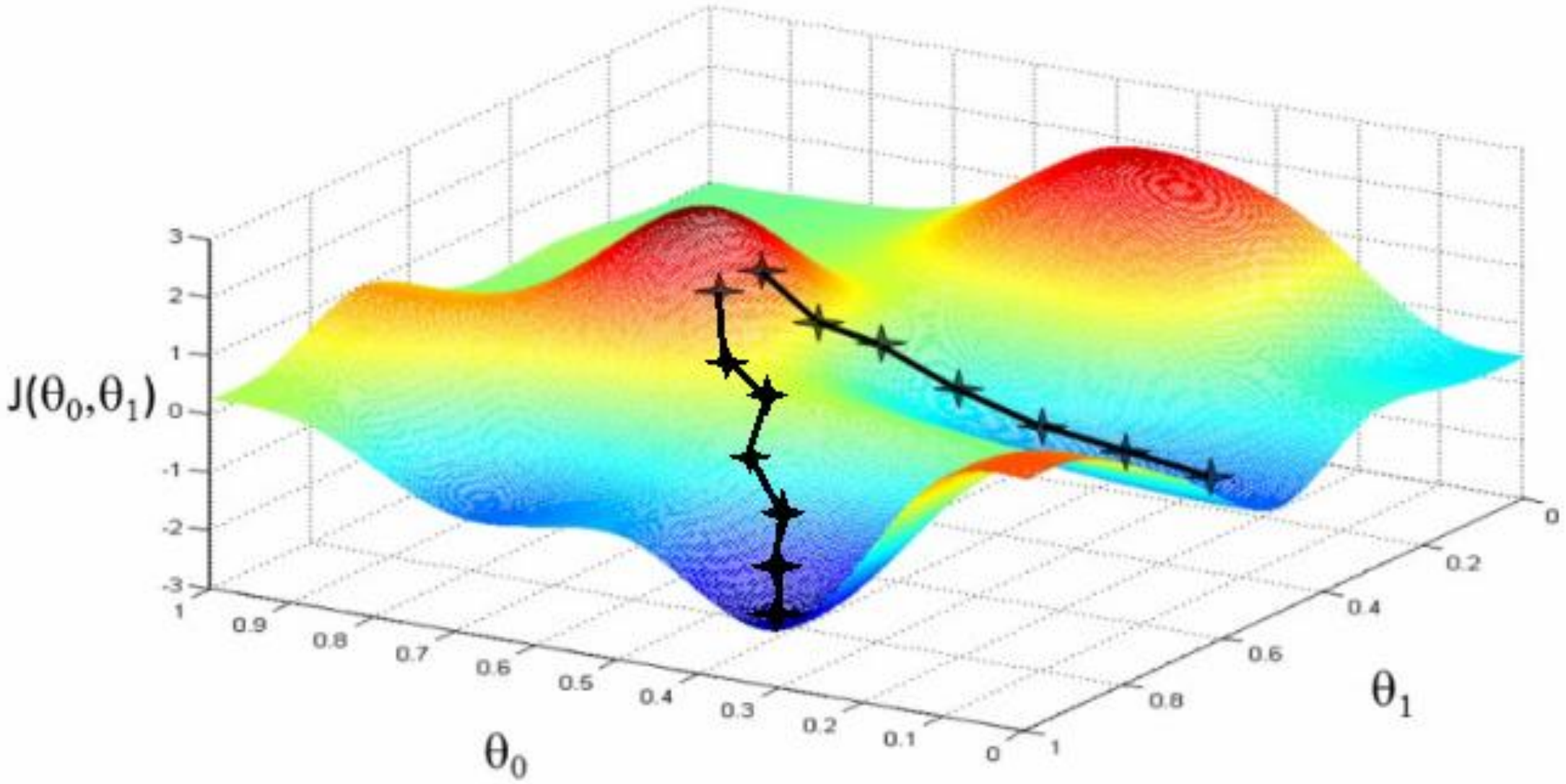
$$L = -\sum_{k=1}^{K} y_k \log(p_k) \qquad \mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Un réseau de neurone est défini par ses poids

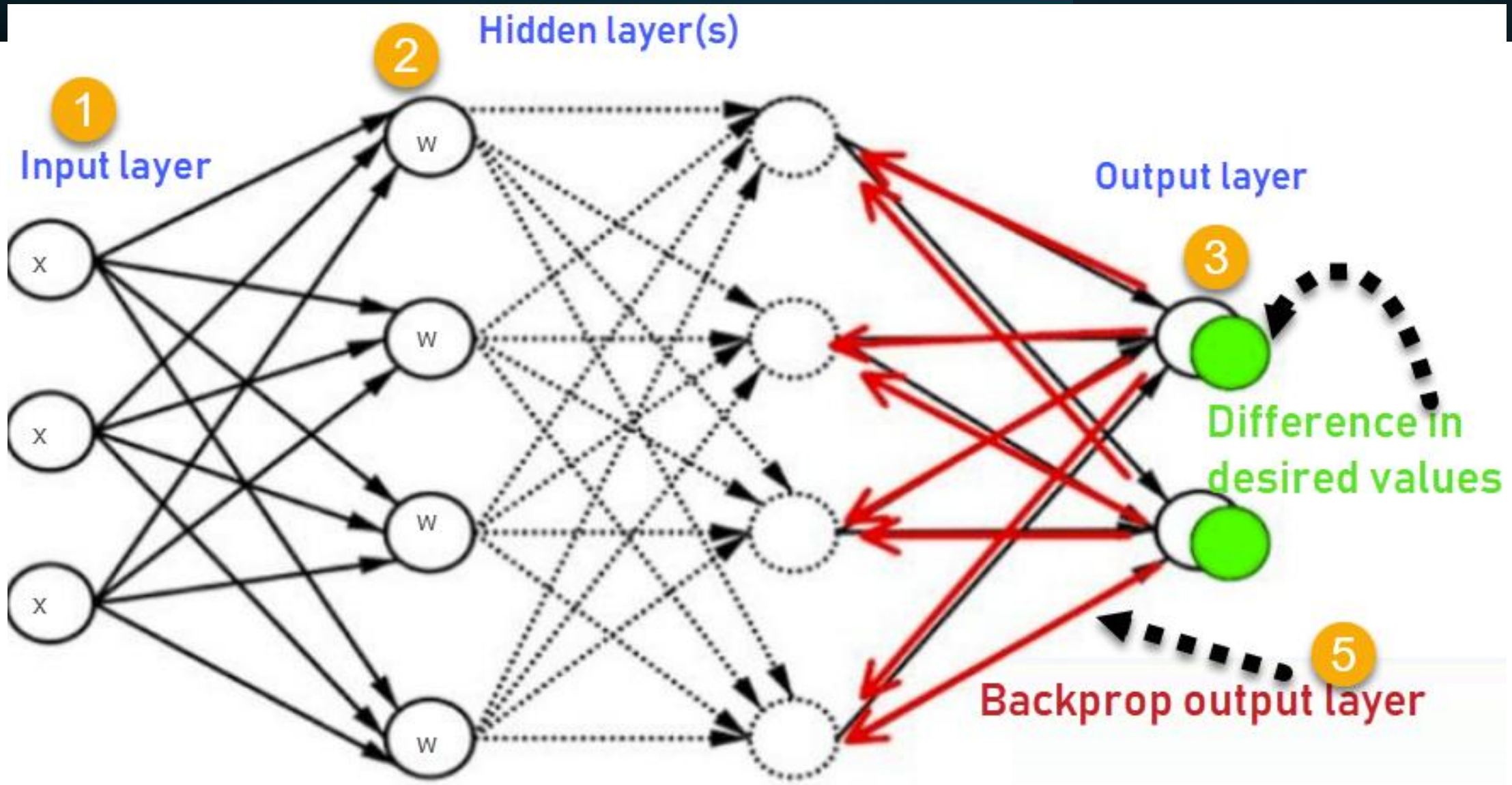


Algorithme de descente de gradient




$J(\theta_0,\theta_1)$
3
2
1
0
-1
-2
-3
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
θ_0
0
0.2
0.4
0.6
0.8
1
θ_1

En pratique : la backpropagation

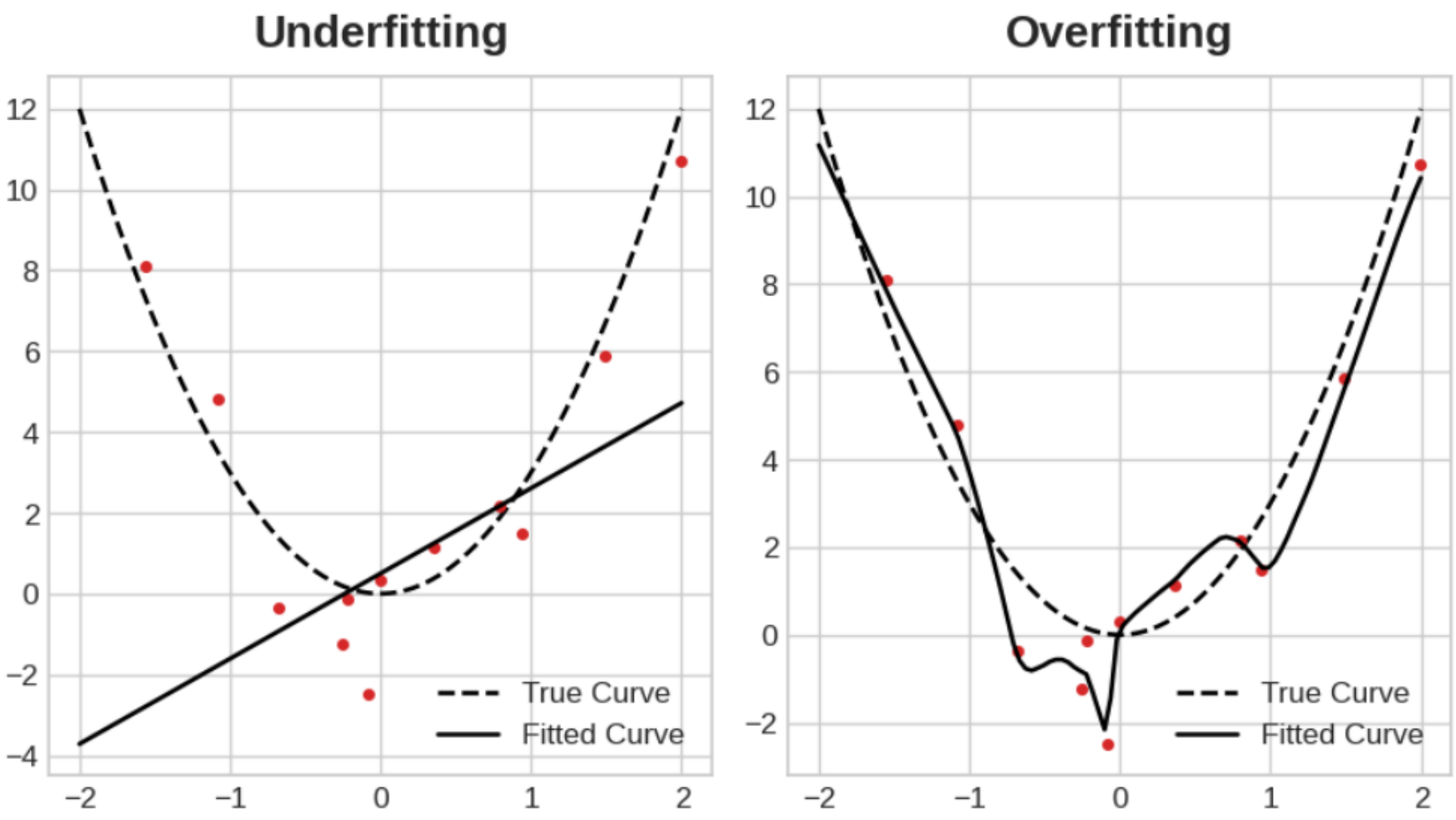


Fitted Line
10
5
0
−5
y
−2
0
2
x
Training Loss
40
30
20
10
0
0
50
100
150
Batches Seen
Weights
4
3
2
1
0
−1
−2
W
b
0
50
100
150
Batches Seen

y = Wx+b

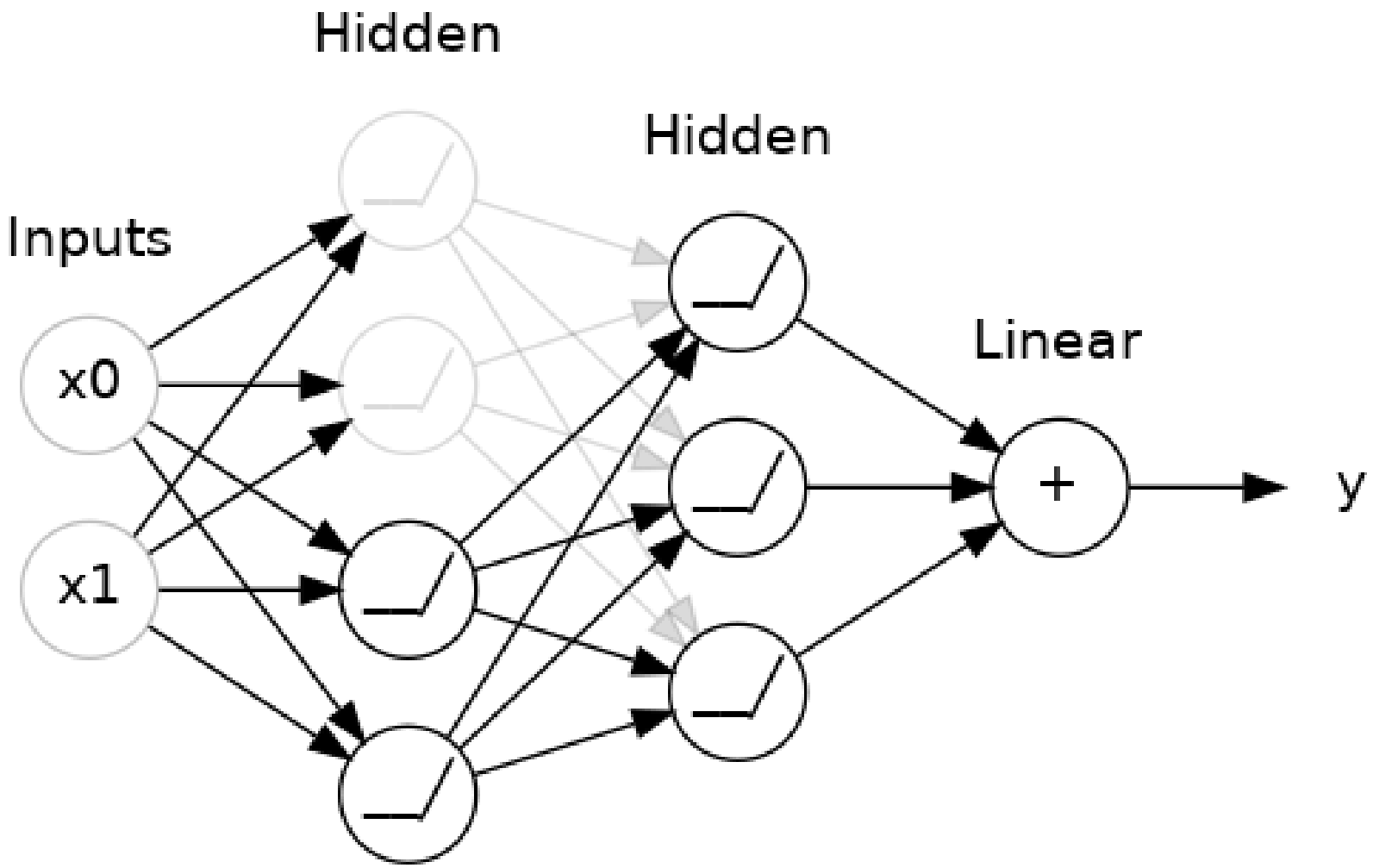
D’autres types de layer

Underfitting vs Overfitting

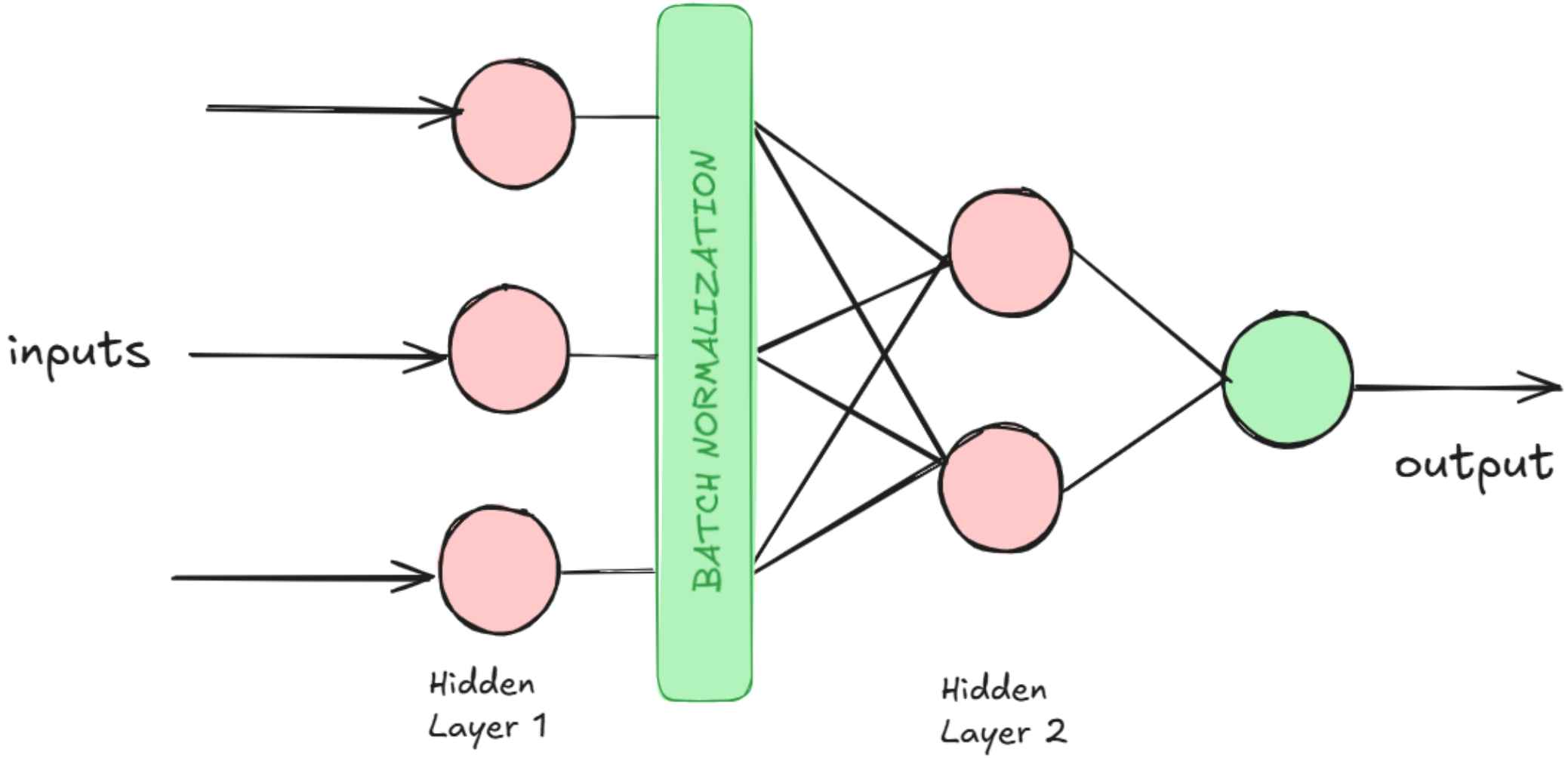


Underfitting and overfitting.

Dropout



Btach Normalization



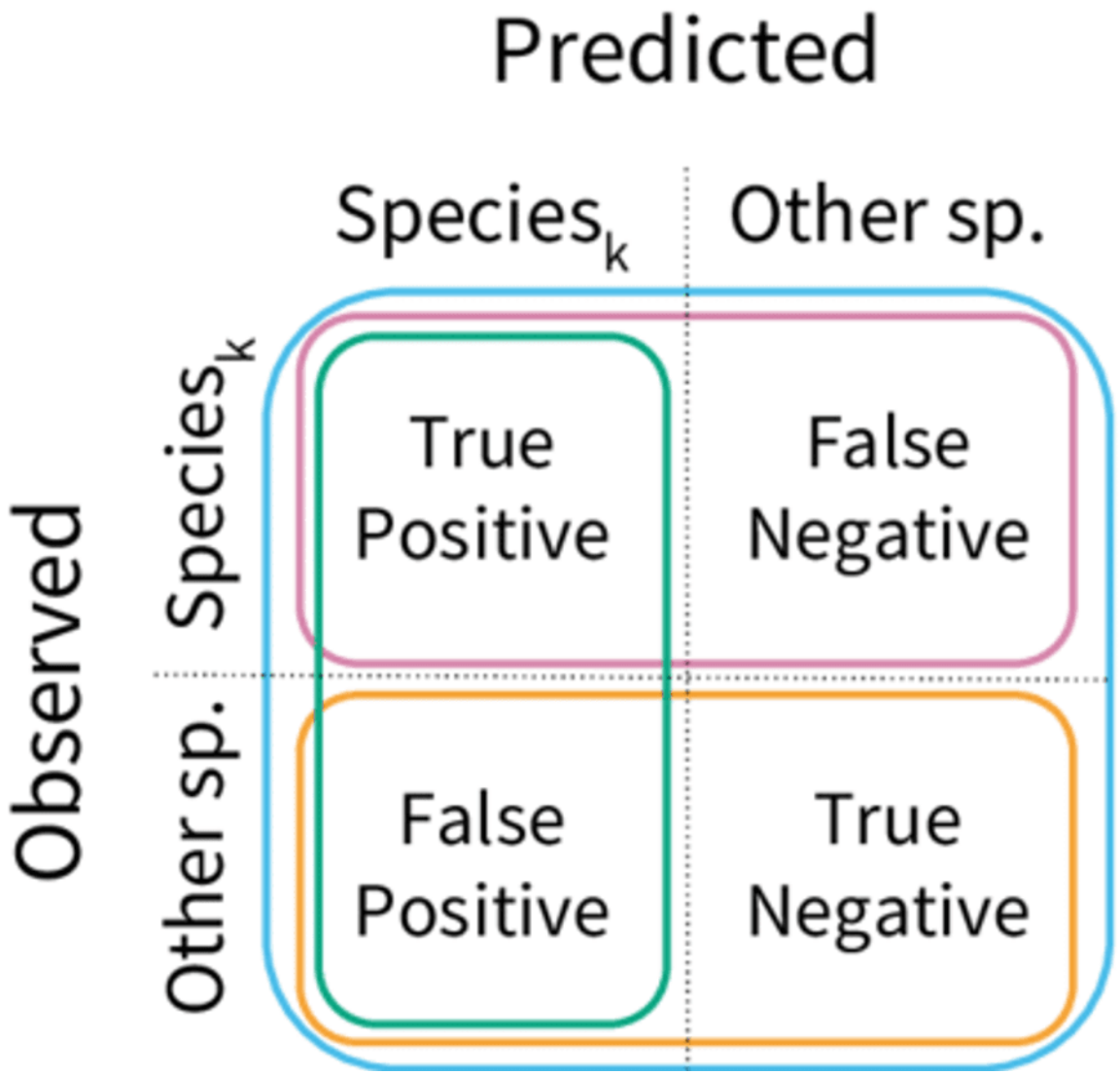
Evaluer un modèle

Evaluer un modèle de régression : mean squared error

$$\mathrm{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{y}_i)^2$$

Evaluer un modèle de classification : accuracy

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}}$$



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$